

Analysis of Seasonal Chlorophyll Variation in the Lower Benguela Region of South Africa

Core 3.4 Python Applications in Ocean Sciences

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Introduction to the Importance of the South Western Cape Upwelling Region

The South Western Cape is one of the sites that forms part of the Benguela Current Marine Ecosystem: a western boundary large marine ecosystem along the coast of south-western Africa. The entire Benguela Current Large Marine Ecosystem stretches from 34°S, south of the Cape Peninsula, northwards to 15°S, at the Namibian Angolan border. Inshore of the Bengula Current proper the prevailing south-easterly winds drive the upwelling associated with the eastern boundary current.

Important upwelling cells within the region were identified by Shannon 1985 as the Cape Frio (19°S), Walvis Bay (23°S), Lüderitz (25–26°S), Namaqua (30°S), Cape Columbine (32°S), and Cape Town (34°S) upwelling cells, each of which form distinct ecoregions due to unique oceanographic conditions. Upwelling events vary seasonally such that maxima in phytoplankton biomass emerge in mid-late summer and early autumn (Martin *et al* 2024). It is hoped that this project may reveal this established upwelling pattern.

Prominent headlands act as a bottleneck for winds, causing them to speed up, thereby enhancing upwelling (Jury 1985). The feature is a distinct shelf-edge jet, known as the Cape Columbine jet that transports water northwards, extending from the Cape Peninsula to Cape Columbine (Veitch *et al* 2018). The area creates a tongue of cold newly upwelled water that often feeds into the St Helena Bay region, fueling intense plankton blooms (Johnson and Nelson 1999).

Datasets and Metadata

Bathymetry Dataset

Description:

Bathymetric data for the lower Benguela Current region were obtained from the Global Multi-Resolution Topography (GMRT) map application tool. The dataset provides gridded seafloor elevation data used to visualize the continental shelf and offshore bathymetric gradients within the study region. The file was downloaded in Coords/CF Compliant NetCDF format in the lowest grid resolution of 489 n/mode. It was then processed and analyzed using rasterio in Python.

Metadata:

```
{'driver': 'netCDF', 'dtype': 'float64', 'nodata': nan, 'width': 716, 'height': 636, 'count': 1, 'crs': None, 'transform': Affine(0.004400677447552681, 0.0, 15.910397317526224, 0.0, -0.0036653003896844893, -32.44119800806938)}
```

The bathymetry dataset was stored in a NetCDF format with raster dimensions of 716 x 636 grid cells, covering the lower Benguela Current region. Missing values were represented as NaN values. Although the coordinate reference system (CRS) was not defined within the metadata, the geographic positioning and longitude/latitude extents indicate that the dataset uses geographic coordinates.

Chlorophyll Dataset

Description:

Chlorophyll-a concentration data were obtained from the ESA-CCI Ocean Color monthly climatology dataset at 4 km spatial resolution. Chlorophyll-a concentration is commonly used as an indicator of phytoplankton biomass and ocean productivity. The dataset was analyzed in Python using xarray.

Metadata:

```
<xarray.DataArray 'chlor_a' (time: 12, lat: 4320, lon: 8640)> Size: 2GB
[447897600 values with dtype=float32]
Coordinates:
  * time      (time) datetime64[ns] 96B 1998-01-01 1998-02-01 ... 1997-12-01
  * lat       (lat) float64 35kB 89.98 89.94 89.9 89.85 ... -89.9 -89.94 -89.98
  * lon       (lon) float64 69kB -180.0 -179.9 -179.9 ... 179.9 179.9 180.0
Attributes:
  standard_name:      mass_concentration_of_chlorophyll_a_in_sea_water
  long_name:          Chlorophyll-a concentration in seawater (not log-tr...
  units:              milligram m-3
  grid_mapping:       crs
  ancillary_variables: chlor_a_log10_rmsd chlor_a_log10_bias
  parameter_vocab_uri: http://vocab.ndg.nerc.ac.uk/term/P011/current/CHLTVOLU
  units_nonstandard:  mg m^-3
```

The chlorophyll dataset consisted of monthly climatological chlorophyll-a concentrations from the ESA Ocean Colour Climate Change Initiative (ESA OC-CCI). The dataset contained 12 monthly time steps with a global spatial grid of 4320 latitude cells and 8640 longitude cells at approximately 4 km spatial resolution. Chlorophyll-a concentrations were stored as 32-bit floating point values and reported in units of milligrams per cubic meter (mg/m^{-3}). The metadata indicates that the dataset uses geographic latitude and longitude coordinates and follows CF-compliant metadata conventions.

Figures

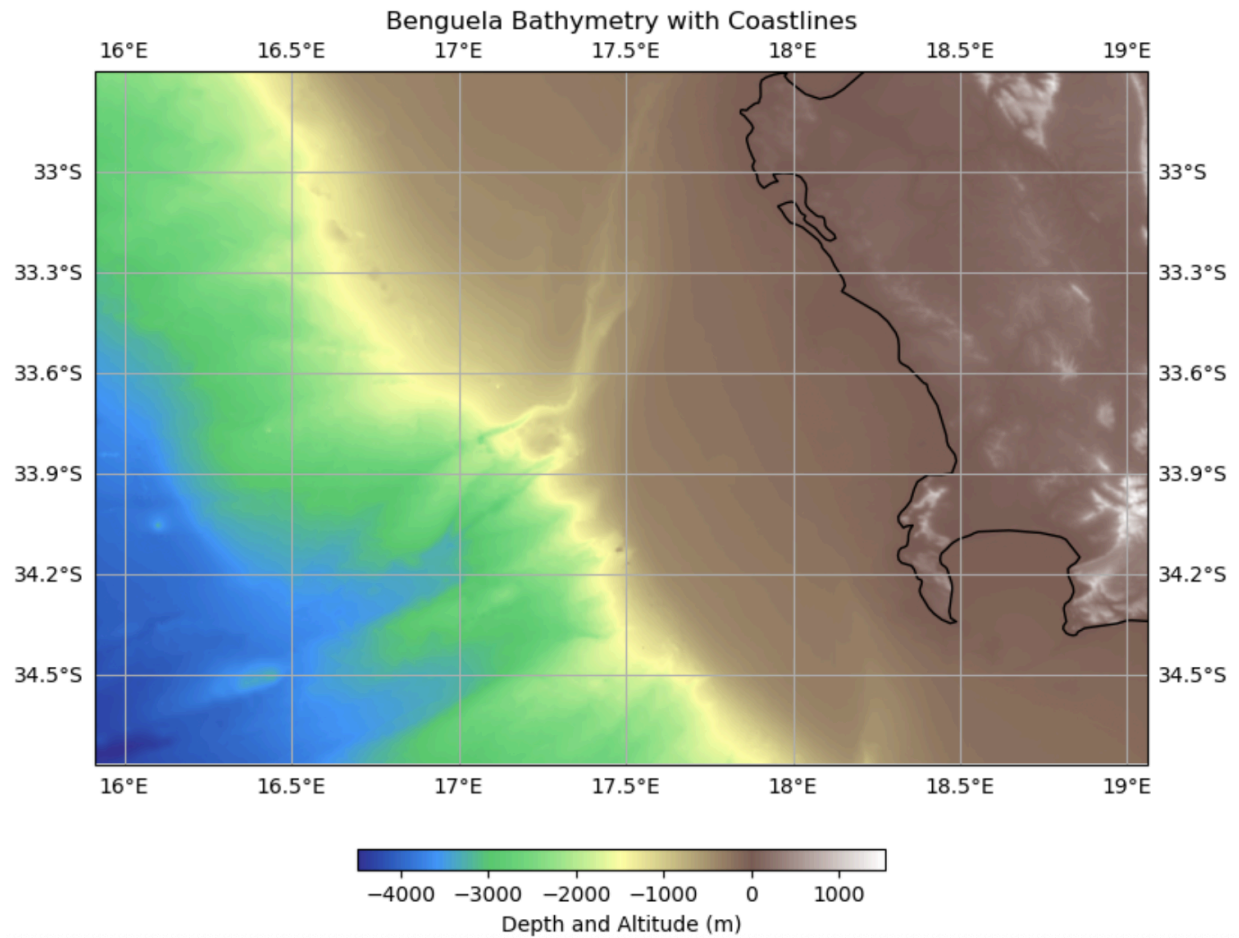


Figure 1: A bathymetric profile of the Southern Benguela shelf off the coast of South Africa.

The bathymetric map shows a narrow continental shelf along the southwestern Cape, which transitions rapidly into a steep continental slope. The proximity of the 200 m isobath to the coastline indicates a compressed shelf system. This bathymetric setting enhances coastal upwelling by increasing the efficiency of wind-driven Ekman transport, allowing nutrient-rich subsurface waters to reach the surface more rapidly. The steep shelf break (200 m isobath) represents a major transition zone between the continental shelf and the deep South Atlantic, where strong bathymetric gradients influence nearshore circulation and the spatial expression of upwelling.

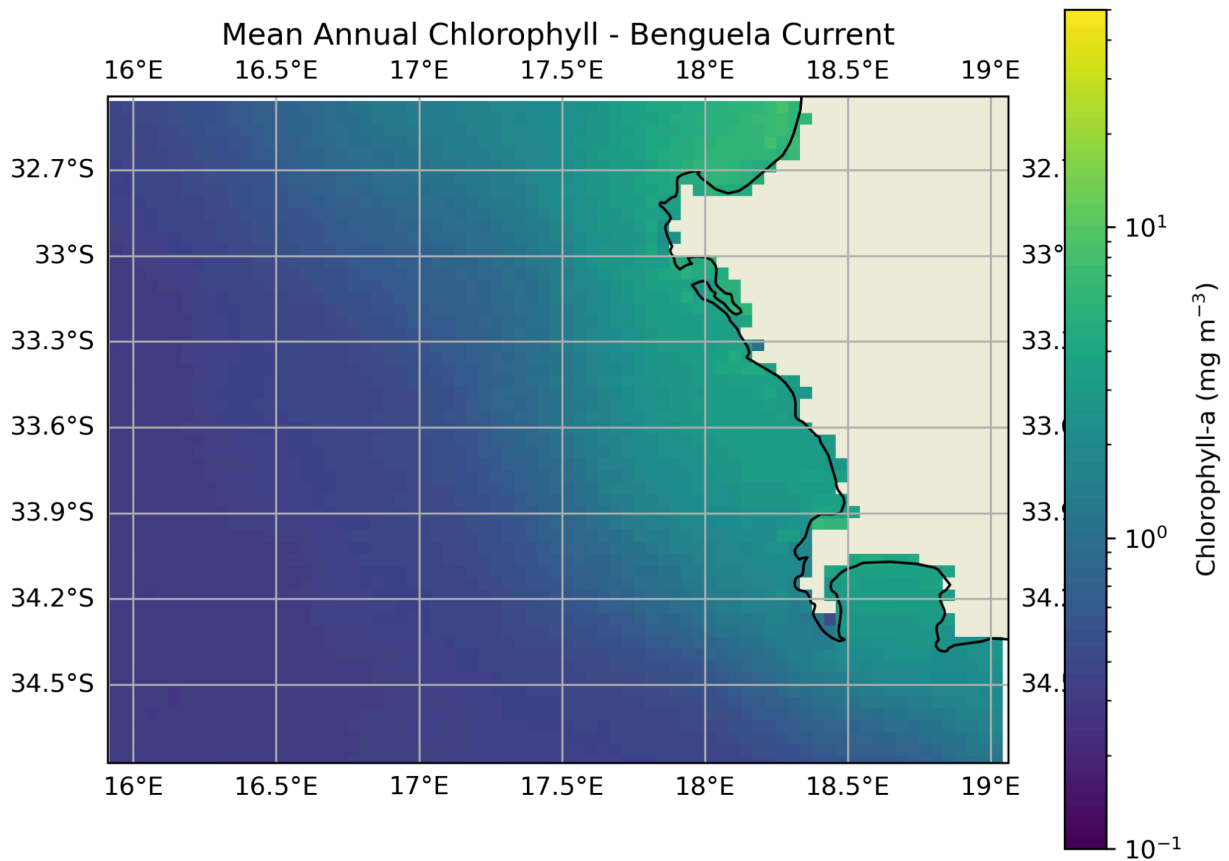


Figure 2: The mean annual surface chlorophyll-a concentration in the Benguela region of South Africa.

The mean annual chlorophyll-a concentrations show a clear inshore–offshore gradient, with the highest concentrations occurring along the continental shelf. This reflects a sustained nutrient supply from coastal upwelling, which supports high phytoplankton biomass. Offshore waters show much lower chlorophyll-a concentrations, typical of oligotrophic conditions in the open South Atlantic.

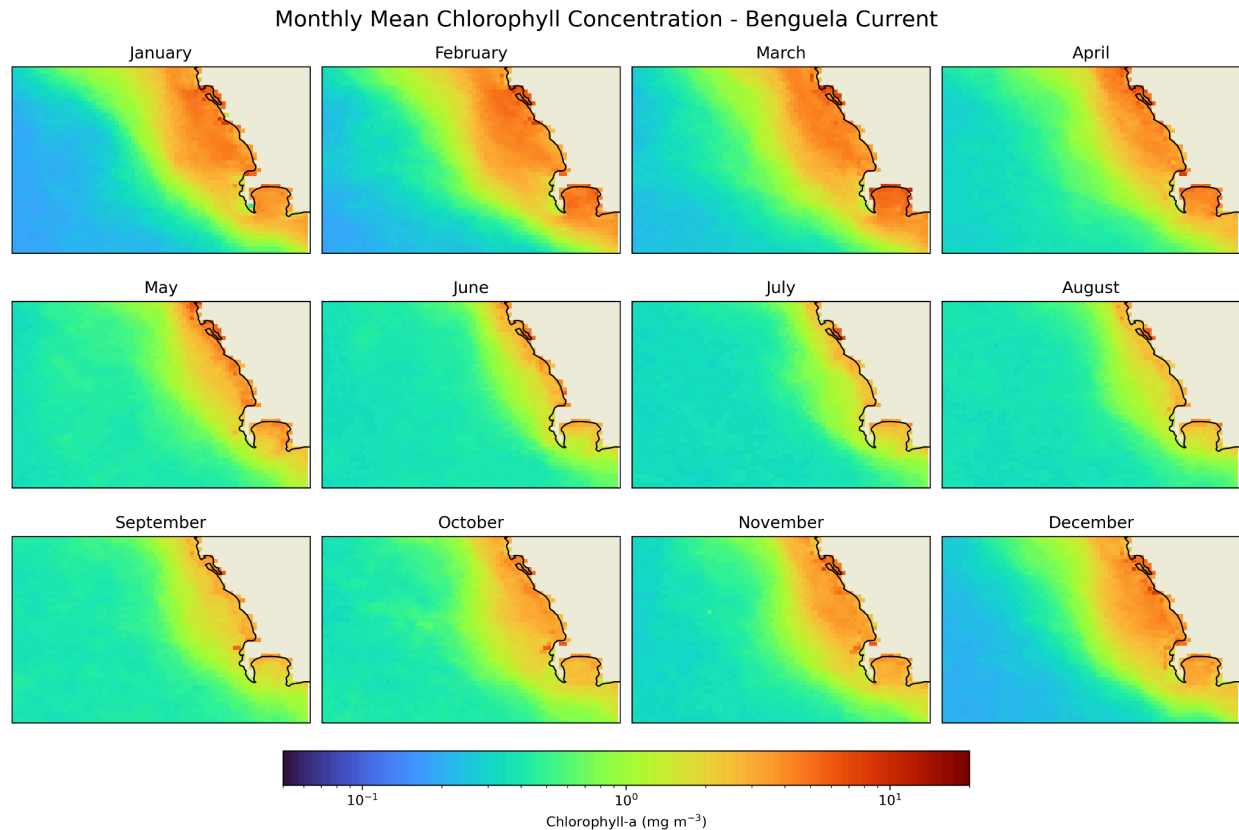


Figure 3: Monthly climatology of chlorophyll-a in the Benguela current region of South Africa.

The monthly chlorophyll-a patterns show strong seasonal changes linked to wind-driven upwelling. During spring and summer (December-March), stronger south-easterly winds enhance offshore Ekman transport, increasing coastal upwelling and leading to higher chlorophyll-a concentrations along the coast. During winter (May–August), weaker upwelling-favourable winds and increased mixing winds lead to deeper mixed layers and reduced surface chlorophyll-a levels. As a result, the productive zone becomes smaller and less intense compared to summer months.

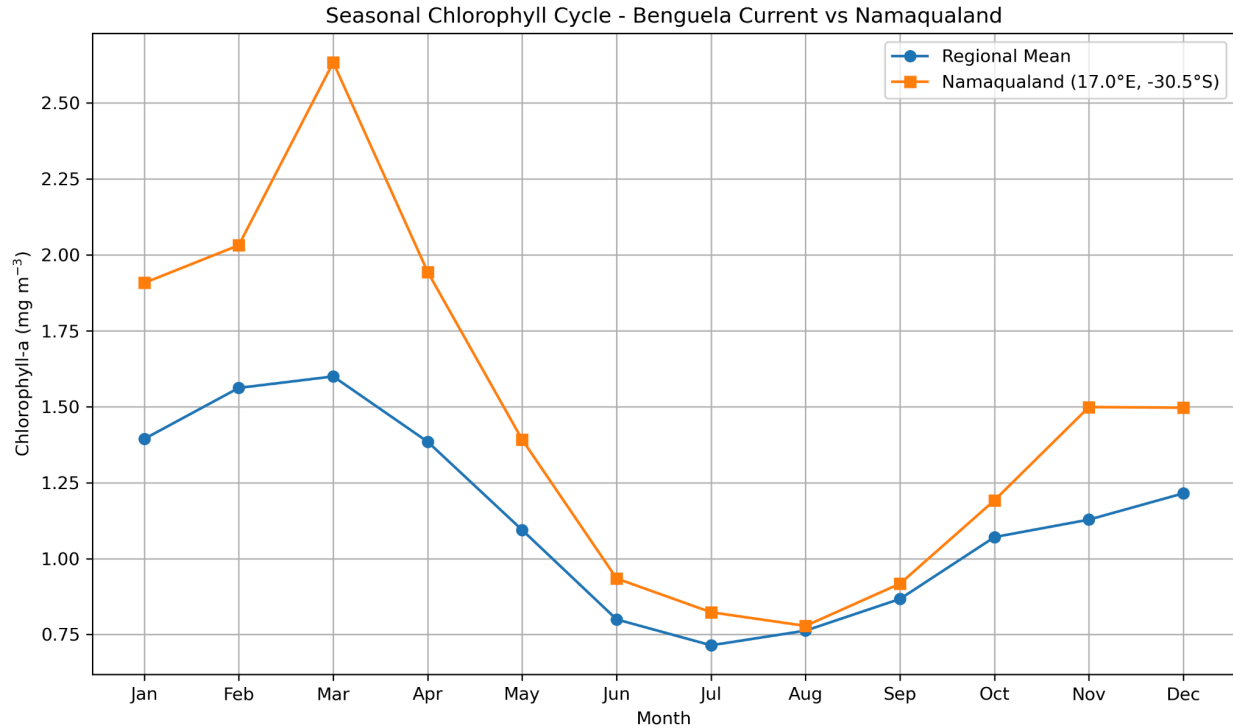


Figure 4: Comparative time series of the regional and Namaqualand upwelling cell seasonal cycles.

The time series comparison shows differences between large-scale and local variability in chlorophyll-a. Chlorophyll-a concentration at Namaqualand is almost double the regional mean because there is a strong upwelling centre located offshore of this region. This results in a more consistent supply of nutrient-rich subsurface water, which supports higher phytoplankton biomass. In contrast, the regional mean represents an average across areas with varying levels of productivity, including both highly productive coastal zones and less productive offshore waters. As a result, the regional signal appears smoother and less variable. The Namaqualand time series also shows greater short-term fluctuations, reflecting episodic upwelling relaxation cycles driven by changes in wind stress. These events lead to rapid increases in chlorophyll-a following nutrient input, highlighting the strong link between physical forcing and biological response at local scales.

Disclaimer on the Use of AI

Artificial intelligence tools (ChatGPT, Copilot) were used in the development of this assignment as a support tool for coding, debugging, and resolving issues. Specifically, AI assistance was used to troubleshoot Python errors related to xarray, cartopy plotting, and GIT workflow issues, as well as to refine figure formatting and improve visualization. AI was also used to help interpret dataset metadata to improve the clarity and structure of scientific writing.

Example prompts used include:

- “Why am I getting TypeError: No numeric data to plot in xarray?”
- “How do I adjust the placement of my chlorophyll maps to add space between them?”
- “How do I move the colorbar below the figure, rather than it being placed on top of the maps?”
- “Why is my GIT push failing due to this error”

All scientific interpretation and final figure selection were checked and adjusted by the author.

References

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